

Chapter 140 — Retroperitoneal Tumours

OVERVIEW

- The **retroperitoneum (RP)** is a common site of pathology relevant to urologists; its tumours frequently involve the **urinary tract, retroperitoneal vasculature and lymphatics**.
- Tumours may be **primary to the RP** or **metastatic deposits** that use it as a landing zone; distinguishing malignant from benign and primary from metastatic drives management.
- Conditions unique to the kidney and adrenal are covered in their own chapters.

ANATOMIC CONSIDERATIONS OF THE RETROPERITONEUM

- **Borders:** diaphragm superiorly; inlet of the true pelvis inferiorly; peritoneal contents anteriorly; posterior body wall posteriorly; flank musculature laterally.
- **Contents:** upper urinary tract (kidneys, renal pelvis, proximal and mid ureter); adrenals; great vessels (infradiaphragmatic aorta and vena cava, common iliac vessels, lumbar branches); RP lymphatics and connective tissue (RP fat, Gerota's fat and fascia); sensory, motor and autonomic nerves (sympathetic chain of ganglia, postganglionic sympathetic and parasympathetic fibres).
- **Adjacent peritoneal organs** — second and third parts of the duodenum, ascending and descending colon, pancreas, splenic hilum — demand attention because RP tumours are **expansile and distort or invade** their neighbours.

RIGHT VERSUS LEFT RETROPERITONEUM — STRUCTURES AT RISK

Region	Structures of greatest concern	Assessment note
Right upper RP	Kidney, ureter, adrenal; more importantly duodenum, inferior vena cava (IVC), posterior diaphragm, right liver segments	Right colonic mesentery hard to assess; indistinct borders and loss of fat planes suggest involvement
Left RP	Aorta, splenic hilum, tail of pancreas; diaphragm; left colonic mesentery	As for right
Both sides	Posterior body wall and psoas	Depth of psoas invasion predicts nerve injury

- **Genitofemoral nerve injury** → transient or permanent **paraesthesia of the anterior and medial thigh**.
- **Psoas and femoral nerve injury** → **motor weakness of hip flexion and knee extension**, with functional impairment — the greater concern.
- **Preoperative anticipation** of involved structures gives a safer operation; consult hepatobiliary, vascular, thoracic and/or general surgeons as the tumour's location requires, and set realistic expectations of convalescence.

KEY POINTS

- The RP is a discrete compartment that houses or borders parts of the urinary, vascular, hepatobiliary, gastrointestinal, neurological and lymphatic systems.
- Its tumours therefore often involve several organ systems and require careful planning before any intervention.

Concept — The RP is defined by what surrounds it; because its tumours grow silently and expansively, the operative problem is usually the **neighbouring structure**, not the tumour itself.

INITIAL EVALUATION OF THE RETROPERITONEAL MASS

- **Silent growth:** the insulation of the RP lets tumours reach considerable size before symptoms; resected RP sarcomas average **more than 15 cm**.
- Whether found incidentally or through symptoms, the next steps follow the same sequence: imaging → serology → tissue.

Imaging

Modality	Role	Limitation
Computed tomography (CT) of chest, abdomen and pelvis	Optimal initial imaging (National Comprehensive Cancer Network (NCCN) guidelines): size, internal density, calcification, relation to organs, border infiltration, regional and distant spread. Intravenous contrast shows cystic change and necrosis; oral contrast shows duodenal, small bowel and colonic involvement. Negative Hounsfield units (fat) can be pathognomonic — liposarcoma, angiomyolipoma, myelolipoma	Cannot determine histological origin or malignant potential from appearance alone
Magnetic resonance imaging (MRI)	Degree of vascular involvement when central vessels are distorted; defines cystic and fatty components; internal detail when contrast is contraindicated (allergy, poor renal function)	—
Positron emission tomography (PET)-CT	Selective: viability of post-chemotherapy residual masses in metastatic seminoma ; grade and subtype of RP liposarcoma; staging and treatment adaptation in some lymphomas	False positives

Laboratory Testing

- **Young man with an RP mass → alpha-fetoprotein (AFP) and human chorionic gonadotrophin (hCG):** critical for diagnosis, prognosis and management of germ cell tumours (GCTs).
- **Blood counts** identify dyscrasias suggesting lymphoproliferative disease; **metabolic panels** assess renal and hepatic function.

Tissue Sampling

- **Fine-needle aspiration (FNA) or core-needle biopsy** gives histological subtype and grade, and matters most where it **changes the sequence of therapy**.
- **Lymphoma:** initial therapy is **systemic chemotherapy in all cases**, regimen by subtype — biopsy is essential.
- **Sarcoma:** surgery was historically first, so biopsy played little role; high-grade tumours are now often treated multimodally, so **biopsy selects patients for up-front chemotherapy, immunotherapy, targeted agents or radiotherapy**.
- Route (transperitoneal vs retroperitoneal) and type (FNA vs core) follow presentation and access; **open or laparoscopic biopsy** when larger samples are needed, usually in advanced or unresectable disease.
- **Core (14- or 16-gauge coaxial) over FNA:** more adequate tissue and preserved cellular architecture.
- **Exception — liposarcoma:** biopsy is often **non-diagnostic** and of limited value in guiding treatment.

KEY POINTS

- Many RP tumours grow silently over months or years and present large, with adjacent organ involvement.

- Cross-sectional imaging best assesses size and extent; serology is important, specifically in GCTs.
- FNA and/or core biopsy should be considered in the initial work-up of every RP mass.

Concept — Imaging defines **extent but not identity**; identity comes from markers and tissue, and biopsy earns its place exactly where the result would change the first treatment.

DIFFERENTIAL DIAGNOSIS

- RP tumours fall into **primary and metastatic tumours of germ cell/gonadal, mesodermal, neurogenic and haematological origin**; management differs widely between groups.

TABLE 140.1 IMMUNOHISTOCHEMICAL AND GENETIC CHARACTERISTICS

Entity	Notable positives	Notable negatives	Genetic markers
Germ cell tumours			
Seminoma	Nuclear: OCT4, SALL4, NANOG. Cytoplasmic: PLAP, CD117, D2-40	CD30, pan-keratin	i(12p)
Nonseminoma (all)	—	Inhibin, calretinin	i(12p)
Embryonal carcinoma	Nuclear: OCT4, SALL4, SOX2. Cytoplasmic: CD30, pan-keratin, CK7	CD117, D2-40	
Yolk sac tumour	AFP, SALL4, glypican-3, pan-keratin	OCT4, CD30, CK7	
Choriocarcinoma	hCG, pan-keratin	OCT4, CD30	
Teratoma	No distinguishing entry; diagnosed on architecture		
Sex cord stromal tumours			
Leydig, granulosa	Inhibin, calretinin	SALL4, PLAP	
Sertoli	Inhibin, calretinin; nuclear β-catenin*	SALL4, PLAP	Subset: CTNNB1 (β-catenin) mutations
Mesenchymal			
Well-/de-differentiated liposarcoma	MDM2, CDK4, S100		Ring chromosomes or 12q amplification
Pleomorphic liposarcoma	S100	MDM2, CDK4	
Leiomyosarcoma	SMA, desmin, h-caldesmon	MDM2, CD117, HMB-45, DOG1	
Undifferentiated pleomorphic sarcoma (UPS)†	Variable; usually unhelpful	Lineage markers	
Synovial sarcoma	TLE1, CD99, EMA, pan-keratin	FLI1	t(X;18)
Solitary fibrous tumour	STAT6, CD34	Desmin, S100, DOG1	NAB2-STAT6 fusion
Extraskeletal Ewing sarcoma	CD99, FLI1	TLE1, EMA, pan-keratin	EWSR1 translocations, especially EWSR1-FLI1, t(11;22)
Desmoplastic small round cell tumour	EMA, desmin, pan-keratin, WT1 (C-terminus antibody only)		t(11;22)(p13;q12) WT1-EWSR1

Entity	Notable positives	Notable negatives	Genetic markers
PEComa	Variably melanocytic (HMB-45, Melan A) and muscle (SMA, desmin)		TSC2 deletions; subset TFE3 rearrangements
Desmoid (fibromatosis)	Nuclear β -catenin		Gardner syndrome (APC); sporadic CTNNB1
Neurogenic			
Ganglioneuroblastoma	CD56, chromogranin A, neurofilament, synaptophysin, neuron-specific enolase		Loss of heterozygosity 11q, 1p; MYCN amplification
Paranglioma	Synaptophysin, chromogranin A, GATA3		Neurofibromatosis type 1 (NF-1); SDHB
Neurofibroma	S100		NF-1 (multifocal and/or plexiform)
Malignant peripheral nerve sheath tumour (MPNST)	S100 (patchy)	H3K27me3 (lost in about half)	NF-1 when from neurofibroma or major nerve; subset PRC2 inactivation
Haematological and other			
Classic Hodgkin lymphoma	CD30, PAX5	CD45	
Non-Hodgkin lymphoma	Variable; usually CD45		
Plasma cell neoplasm	CD138, CD38	CD45, CD20	
Cystic lymphangioma	CD31, CD34, D2-40		
Cystic mesothelioma	Calretinin, pan-keratin		
RP cancer of unknown primary	No distinguishing entry		

* Nuclear β -catenin is very specific for Sertoli cell tumour but seen in only a majority; its absence does not exclude the diagnosis. † Text and table differ for UPS — see that section.

- **What the table is for:** resolving a mass that imaging cannot type. **Pattern:** germ cell tumours sort on OCT4/CD30/SALL4; sex cord stromal tumours on inhibin/calretinin with SALL4/PLAP negativity; mesenchymal tumours on one defining lesion (MDM2 amplification, a translocation or a fusion).
- **Traps:** Hodgkin lymphoma is **CD45-negative**; pleomorphic liposarcoma is **MDM2/CDK4-negative** unlike its well- and de-differentiated relatives; Ewing sarcoma and desmoplastic small round cell tumour are **both t(11;22) with EWSR1** but fuse to different partners (*FLI1* vs *WT1*).

Germ Cell/Gonadal Origins

- Primary in the RP or metastatic to it. Uncommon overall, but an **RP mass in a young man must immediately raise the question of GCT**.
- Sex cord stromal tumours of the testis also use the RP as a predictable landing zone. Seminoma, nonseminoma and the sex cord stromal tumours (Leydig, Sertoli, granulosa) each have distinct biology dictating management.

Germ Cell Tumours

- Covered fully in Chapters 79 and 80; relevant here for their **propensity to metastasise to the RP** and occasional primary RP origin.
- **Incidence rising**, most rapidly in Hispanic men. Usually present as an enlarging scrotal mass; in the RP, as **heterogeneous, circumscribed masses along the RP lymph node chains**.

- **CT** best characterises RP disease; **scrotal ultrasound (US)** detects the testicular primary.
- Seminoma and nonseminoma differ in distribution and rate of distant disease, serology, sensitivity to chemotherapy, surgery or radiotherapy, and long-term relapse risk.

SEMINOMA VERSUS NONSEMINOMA

	Seminoma	Nonseminoma
Origin, components	Germinal epithelium of seminiferous tubules; about half of GCTs at presentation	At least one of embryonal carcinoma, yolk sac tumour, choriocarcinoma, teratoma
Age	Third to fifth decades	Late teens to early 30s (later possible)
Markers	hCG raised in 10–15% (syncytiotrophoblasts); tumour-attributable AFP excludes pure seminoma	AFP: yolk sac and embryonal. hCG: choriocarcinoma, embryonal (and seminoma)
Gross, histology	Fleshy, tan; sheets of polygonal clear cells in nests divided by fibrous septa; lymphocytic infiltrate in most; uniform, orderly prominent nucleoli ; numerous mitoses	Variable haemorrhage, necrosis, cystic change, calcification
Subtypes	Classic, spermatocytic (rarely if ever metastasises), anaplastic (classic vs anaplastic has little prognostic value)	Mixed components
Treatment sensitivity	Mostly confined and cured by orchidectomy ; metastatic disease chemo- and radiosensitive	Embryonal, yolk sac and choriocarcinoma chemosensitive; teratoma chemo- and radioresistant → surgery central in advanced disease
Prognosis	One of the most curable solid tumours; metastatic 5-year overall survival (OS) 95% good-risk, 88% intermediate-risk (International Germ Cell Cancer Collaborative Group (IGCCCG) update)	—

NONSEMINOMATOUS COMPONENTS

Component	Histology	Immunohistochemistry (IHC)
Embryonal carcinoma	Clusters of polygonal cells, variably eosinophilic cytoplasm, prominent nucleoli, disorganised (contrast seminoma)	Nuclear OCT4, SALL4, SOX2; CD30, CK7, pan-keratin in most; AFP in up to a third
Yolk sac tumour	Variable architectural pattern	SALL4, glypican-3, pan-keratin positive; weakly AFP; OCT4 and CD30 negative
Choriocarcinoma	Extensive haemorrhage and necrosis ; syncytiotrophoblasts and mononuclear trophoblasts	Diffuse hCG, pan-cytokeratin; OCT4 and CD30 negative
Teratoma	Random elements of several germ layers — squamous/glandular epithelium, smooth muscle, bone, cartilage, mature and immature neural tissue; cystic change common; somatic transformation into sarcoma or carcinoma	Little role — architecture and marbling are diagnostic

- **Chromosome 12p abnormalities** (isochromosome 12p) are characteristic in most GCTs.
- **Extragenital GCT**: GCT confined to mediastinum or RP without testicular primary accounts for **3–5% of GCTs**; the RP is one of the two principal sites. True incidence is **controversial**: when an RP

"extragonadal" GCT lateralises to the left or right landing zone, **most patients have teratoma or scar (necrosis/fibrosis) in the ipsilateral testis** at post-chemotherapy orchidectomy — a burnt-out primary.

- **Management:** GCT identified serologically or pathologically in the RP → **platinum-based chemotherapy first**; if nonseminomatous → **resect residual disease after chemotherapy in all cases.**

Concept — Every management difference between seminoma and nonseminoma flows from **teratoma**: chemoresistant and marker-silent, it makes surgery central for nonseminoma and marginal for seminoma.

Sex Cord Stromal Tumours

- **Less than 5% of testicular neoplasms**; Leydig, Sertoli and granulosa cell types.

	Leydig cell	Sertoli cell	Granulosa cell
Frequency	Commonest sex cord stromal tumour	<1% of testicular neoplasms	Rarest ; juvenile type up to 7% of prepubertal testicular tumours
Age	30–60 years	35–50 years	Juvenile: mostly before 6 months ; adult type
Hallmark	Reinke crystals — classic but present in only about 30%	Association with Peutz-Jeghers syndrome	Juvenile: lobular growth with variably sized follicles, benign . Adult: resembles ovarian granulosa tumour; a quarter present with infertility and gynaecomastia ; slow growing, RP metastasis rarely reported
Differential	Tumours of congenital adrenal hyperplasia → distinguished by raised adrenocorticotrophic hormone (ACTH)	Seminoma → cytokeratin and inhibin positive, placental alkaline phosphatase (PLAP) negative favours Sertoli	—

- Most are benign, but **about 10% metastasise**; malignant forms **respond poorly to chemotherapy and radiotherapy**. Spread: **RP first, then lung**, as with mixed GCTs.
- **Surveillance is hazardous**: mortality is high once metastasis develops on surveillance, even after adequate resection → interest in **immediate retroperitoneal lymph node dissection (RPLND)** in clinically node-negative patients, for staging and to prevent further nodal spread.
- **Metastasis is the only reliable marker of malignancy**. Primary features predicting aggressive behaviour: older age; **size larger than 4–5 cm**; necrosis; **mitoses more than 3–5 per 10 high-power fields (HPF)**; moderate-to-severe atypia; infiltrative margins or invasion of adjacent structures; lymphovascular invasion.
- **Radiologically enlarged nodes** carry a poor prognosis, although some patients with positive RP nodes are cured.

Concept — Because malignant sex cord stromal tumour resists chemotherapy and radiotherapy, **surgery is the only lever**, so it must be used on primary-tumour risk features before nodes become radiologically evident.

Mesodermal Origins

- Soft-tissue tumours. **Sarcomas**: commonest liposarcoma, leiomyosarcoma, malignant fibrous histiocytoma (MFH); less commonly angiosarcoma, rhabdomyosarcoma, myxosarcoma. **Benign**: lipoma, leiomyoma, fibroma, haemangiopericytoma, perivascular epithelioid cell tumour (PEComa), chordoma, myelolipoma, xanthogranuloma, myxoma, gastrointestinal stromal tumour (GIST).

- **Soft-tissue sarcoma is <1% of adult solid cancers; 15–20% arise in the RP.** About 50 subtypes exist, few of them in the RP; overlap in appearance means diagnosis requires **integrated IHC, molecular, cytogenetic and clinical data.**

RETROPERITONEAL SARCOMAS AND MESENCHYMAL TUMOURS AT A GLANCE

Entity	Frequency	Discriminating feature	Principal treatment
Well-/de-differentiated liposarcoma	Commonest, ~60% of RP sarcomas	Fat density; 12q (MDM2/CDK4) amplification	Radical compartment resection
Leiomyosarcoma	Second, ~20%	From vessel wall or smooth muscle; no fat	Negative margins + adjuvant therapy
MFH/UPS	~A quarter of soft-tissue sarcomas	No fat; heterogeneous; older adults	Wide resection
Unclassified sarcoma	Up to 20%	Diagnosis of exclusion; prior radiotherapy in a quarter	Exclude de-differentiated liposarcoma (MDM2)
MPNST	<10%	Nerve origin; NF-1	Surgery; nerve sacrifice
Synovial sarcoma	Uncommon in RP	Calcification; t(X;18)	Grade-dependent
Solitary fibrous tumour	—	NAB2-STAT6; Doege-Potter	En bloc resection, organ-sparing; radiosensitive
Ewing sarcoma	Rare in RP	Small round blue cell; EWSR1-ETS	Chemotherapy-based multimodal
Desmoplastic small round cell tumour	Rare	WT1-EWSR1; peritoneal implants	Multimodal; universally poor
PEComa	—	Melanocytic + muscle markers; TSC2	Excision; mTOR inhibitors
Desmoid	—	No metastasis; β -catenin	Surgery or observation

Liposarcomas

- Malignant, **locally aggressive** tumours of adipocytes with nuclear atypia and hyperchromasia. **Well-differentiated (WD) versus de-differentiated (DD)** depends on the presence of a higher-grade, non-lipogenic component.
- Both present as **painless, often incidental masses**; DD tumours grow rapidly, signalling the more aggressive phenotype.
- **CT:** low-attenuation (fat-density) mass with mass effect on surrounding structures; **internal higher-density areas indicate de-differentiated components** — a single tumour may contain both.
- **WD histological types:** adipocytic (lipoma-like — scant atypical hyperchromatic nuclei among fat lobules, so **core biopsies are often inconclusive**); sclerosing (hyalinised stroma with scattered atypical adipocytes); inflammatory (lymphoid aggregates may mimic **primary lymphoproliferative malignancy**). Diagnosis usually needs histology plus clinical and anatomical context.
- **DD histology:** arises from WD components — atypical adipocytes surrounding fleshy non-lipogenic areas resembling undifferentiated spindle cell sarcoma, myxoid tumour, meningothelial or pleomorphic patterns; heterologous osteosarcoma, rhabdomyosarcoma, leiomyosarcoma or small round cell elements occur.
- **Molecular:** both show overexpression and/or **amplification of 12q genes MDM2 and CDK4** (with S100 positivity); **fluorescence in situ hybridisation (FISH) or IHC for MDM2/CDK4** confirms the diagnosis in limited samples or where stroma or inflammation obscures it.

WELL-DIFFERENTIATED VERSUS DE-DIFFERENTIATED LIPOSARCOMA — BEHAVIOUR

	Well-differentiated	De-differentiated
Growth	Slow, painless	Rapid
Local recurrence	Up to 40%	Similar
Distant metastasis	Uncommon	Up to 20%
5-year cancer-specific mortality	—	30%

- Local recurrence depends on tumour biology and surgery. The impact of **R0 versus R1** margins is debated, but **negative margins optimise outcome**; **radical compartment resection** (function-preserving) improves local recurrence.
- Average time to recurrence is **about 2 years** → **long-term surveillance** is required.

SYSTEMIC THERAPY FOR UNRESECTABLE OR METASTATIC LIPOSARCOMA

Agent	Class	Line	Result
Anthracycline-based regimens	Cytotoxic	First	Modest short-term survival gain
Trabectedin	DNA-binding agent	After anthracycline	Progression-free survival (PFS) benefit, no OS benefit
Pembrolizumab	Programmed death-1 inhibitor	Second line	Approved for high tumour mutational burden without satisfactory alternatives

Pleomorphic Liposarcomas

- Very rare; adults; **extremities more than RP**; also spermatic cord. Fast growing.
- Pleomorphic lipoblasts that **do not overexpress MDM2 or CDK4**; frequent mitoses; variable spindle and epithelioid cells. On needle biopsy they mimic **poorly differentiated carcinoma or renal cell carcinoma**; identifying poorly differentiated lipoblasts is essential.
- **Rapid progression to widespread metastasis, lung commonest.**

Concept — The liposarcoma family sorts on **MDM2**: amplified in well- and de-differentiated tumours (fatty, locally recurrent disease), absent in pleomorphic tumours (systemic disease); within the amplified pair, de-differentiation is what adds distant spread.

Leiomyosarcomas

- Malignant smooth muscle tumours; **second commonest RP sarcoma (~20%)**. Arise from the **media of RP blood vessels**, the muscularis of the collecting system, or other smooth muscle.
- Grow silently until mass effect or obstruction impairs **venous return, urinary drainage or the gastrointestinal tract**. In women, many are **gynaecological in origin**. Often incidental. Rarely arise with **Epstein-Barr virus infection in immunosuppressed patients**.
- **Homogeneous tumours encasing structures** with fleshy bundles; occasional necrosis or cysts. Fascicles of atypical smooth muscle, atypia graded; de-differentiation uncommon.
- **Benign versus malignant** criteria: location (RP vs somatic soft tissue), mitotic rate, nuclear and cellular atypia, necrosis → spectrum **leiomyoma** → **smooth muscle tumour of uncertain malignant potential (STUMP)** → **leiomyosarcoma**.
- IHC: **SMA, desmin, h-caldesmon positive**; MDM2, CD117, HMB-45, DOG1 negative.

- Local and regional recurrence common; distant spread frequent. **Needle biopsy underestimates aggressiveness** → aggressive multimodal approach; **negative margins with adjuvant therapy** (for distant microscopic disease) give the best prognosis.

Malignant Fibrous Histiocytoma/Undifferentiated Pleomorphic Sarcoma

- Fibroblastic and histiocyte-like cells; **fifth and sixth decades**; about a quarter of soft-tissue sarcomas. Often silent and large at detection.
- **Radiology**: heterogeneous soft-tissue mass (central necrosis, haemorrhage, myxoid degeneration); **absence of fat separates it from liposarcoma**; vascular invasion rare; peripheral nodular enhancement of a lobulated mass; **calcification in about 10%**.
- Metastasis to **lung, liver and bone**.
- **Histology**: spindle fibroblasts, round histiocyte-like cells, lymphocytes, foam cells, giant cells; pseudoangiomatoid pattern; plasma cell and lymphocytic infiltrate; fibrous pseudocapsule.
- **Types**: storiform-pleomorphic and myxoid (**high grade**) versus giant cell, angiomatoid and inflammatory (more often low grade).
- **Treatment is surgical** in the absence of metastasis: **wide resection with negative margins** reduces local and distant relapse and improves survival. Adjuvant chemotherapy or radiotherapy is unproven; 5-year OS about half.

UPS IMMUNOPHENOTYPE — TEXT VERSUS TABLE 140.1

Source	Stated profile
Body text	Desmin, epithelial membrane antigen (EMA), CD68, CD99 positive; fusions <i>FUS-ATF1</i> , <i>EWSR1-ATF1</i> , <i>EWSR1-CREB1</i>
Table 140.1	IHC variable and usually unhelpful; negative for lineage markers

- **Reason**: the text's markers and fusions belong to the **angiomatoid subtype**; UPS as a category is defined by **absence of a line of differentiation**, as the table states.

Synovial Sarcomas

- Uncommon in the RP; **all ages**. Arise from kidney, adrenal or other viscera; nerve origin can mimic MPNST.
- Large and asymptomatic; **intratumoral calcification hints at the diagnosis**. Rarely follows prior radiation.
- Mostly encapsulated with cystic degeneration; **herringbone spindle cell pattern**; mitoses vary with grade; glandular, cystic or round cell differentiation causes confusion.
- **TLE1 positive in about 90%** (not exclusive); **t(X;18) in most**.
- Outcome depends on **grade**; favourable: **younger age, diffuse calcification**.

Solitary Fibrous Tumours

- Formerly **haemangiopericytoma**; fibroblastic; occur anywhere; mostly younger patients; very large before symptoms.
- **Doege-Potter syndrome**: rare paraneoplastic **hypoglycaemia** from oversecretion of insulin-like growth factor.
- "Pattern-less" architecture, sometimes myxoid or fatty; high mitotic index and pleomorphism mark aggressive behaviour (about a third). **Near-uniform NAB2-STAT6 fusion** is unique to this tumour.
- Mostly indolent, but **clinicopathological features predict biology poorly** — some benign-looking tumours metastasise. Risk factors: advancing age, high mitotic rate, necrosis, size. Overtly malignant cases do poorly; 8-year OS about 75%.

- **Complete en bloc resection** is standard, **preserving adjacent organs where feasible** (wide margins not routinely needed). **Radiosensitive** → perioperative radiotherapy as adjunct.

Ewing and Ewing-Like Sarcomas

- Primarily bone tumours; malignant **small round blue cell** tumours, rarely in RP soft tissue. **EWSR1 translocation with the ETS transcription factor family**.
- **Young adult White men**; neuropathic pain when adjacent to nerves.
- **Infiltrative margins**, unlike other sarcomas → **surgery alone rarely curative**; standard is **combination chemotherapy within multimodal therapy**, cure rates above 50%.
- **Poor prognosis**: larger size, metastasis, **little necrosis after chemotherapy**; in Ewing-like sarcoma, **CIC rearrangement**.

Desmoplastic Small, Round Cell Tumours

- Translocation-associated small round cell sarcoma: **t(11;22)(p13;q12) WT1-EWSR1**; staining for the fusion product is **diagnostic**.
- **Young men**; **multifocal peritoneal implants**; abdominal pain and distension; heterogeneous enhancement with central necrosis; **ascites typical**.
- **Prognosis universally poor** despite multimodal therapy.

Unclassified Sarcomas

- No defining line of differentiation — **diagnosis of exclusion**, up to 20% of cases; **prior radiotherapy in a quarter**.
- Rapid growth, aggressive course, universally poor prognosis. **Screen for MDM2 to exclude de-differentiated liposarcoma** and avoid misclassification.

Perivascular Epithelioid Cell Tumours

- Family: renal and extrarenal **angiomyolipoma (AML)**, extrapulmonary clear cell tumour, clear cell myomelanocytic tumour, pigmented melanotic tumour, **lymphangioleiomyomatosis (LAM)**, malignant PEComa — hence diverse behaviour.
- **Malignant features**: size **more than 5 cm**; high cellularity and atypia; **mitoses more than 1 per 50 HPF**; necrosis; vascular invasion.
- **Mostly sporadic**; **tuberous sclerosis** accounts for the remainder; **TFE3 rearrangements** in a subset, usually younger patients.
- **Largely chemoresistant** → wide local excision when adverse features are present; unresectable or metastatic → **mammalian target of rapamycin (mTOR) inhibitors** (temsirolimus, everolimus, sirolimus) give responses and prolonged survival.

Desmoid Tumours

- Fibroblastic, locally advanced; rapid growth and **disruption of fascial planes** → **ureteric or bowel obstruction**. Peak **third decade**. Radiologically heterogeneous, hard to distinguish from sarcoma. Spindle cells, extracellular collagen, myxoid matrix.
- **Do not metastasise**, but **recur locally in up to half despite wide resection**.
- **Surgery** for resectable disease; **observation** is appropriate where growth would not change the surgical outcome or cause functional limitation.

Concept — Among RP sarcomas, treatment is dictated less by size than by **how the tumour spreads**: locally recurrent types (liposarcoma, desmoid) are won or lost at the margin, whereas infiltrative or systemic types (Ewing, desmoplastic small round cell tumour, pleomorphic liposarcoma) need chemotherapy because no margin is achievable.

NEUROGENIC ORIGINS

- Ganglion cells, paraganglionic tissue and peripheral nerve sheaths — **about 10% of primary RP tumours**. Sites: sympathetic ganglia **posterior to the great vessels**, the adrenals, the **organ of Zuckerkandl**.
- **Prognosis much better than sarcoma**; many are benign.

Tumours of Sympathetic Ganglia

	Ganglioneuroma	Neuroblastoma	Ganglioneuroblastoma
Nature	Benign ; nerve cells of sympathetic ganglia	Malignant ; primitive neuroblasts of adrenal medulla and sympathetic chain	Features of both
Age, sex	Third–fourth decade; M = F	First decade ; two-thirds adrenal	Children 2–4 years ; adults late second to early fourth decade
Imaging	Well circumscribed; calcification up to 30%	Heterogeneous; coarse calcification in most	Staging: MRI and metaiodobenzylguanidine (MIBG) for bone and local extent
Behaviour	Usually incidental; adrenal origin may secrete catecholamines, vasoactive intestinal peptide or androgens	Mostly metastatic at presentation ; encasement makes local therapy difficult	Abdominal or back pain; metastasis to nodes, bone, brain, liver, skin; more than half metastatic at presentation
Treatment, prognosis	Excellent after local removal	—	Surgery if localised ± adjuvant; neoadjuvant paediatric-regimen chemotherapy recommended ; poor in adults; MYCN amplification adverse

- Ganglioneuroblastoma regimens combine several of cyclophosphamide, cisplatin, ifosfamide, vincristine, doxorubicin, etoposide, 13-*cis*-retinoic acid and carboplatin; diagnosis by histology (IHC as Table 140.1).

Tumours of the Paraganglionic System

- Specialised **neural crest cells** of adrenal medulla and RP ganglia: **adrenal = pheochromocytoma**; **extra-adrenal = paraganglioma**.
- Extra-adrenal paraganglioma: **third–fourth decades, sex-neutral**. Commonest site: **organ of Zuckerkandl** — **anterior to the aorta below the inferior mesenteric artery (IMA) origin, near the bifurcation**.
- Radiology as for pheochromocytoma: **high signal on T2-weighted MRI**. Catecholamine production in **about 40%**.
- Grossly **yellow-grey**; large polygonal pleomorphic cells; **synaptophysin and chromogranin A positive**.
- Necrosis, high mitotic index, atypical nucleoli, infiltrative borders and vascular invasion **poorly predict malignancy**; **metastases in 20–50%**.
- **Management is surgical** — poor response to other therapies. Predisposition: **NF-1 and multiple endocrine neoplasia type 2A (MEN 2A)**.

Tumours of Nerve Sheath Origin

	Schwannoma	Neurofibroma	Malignant peripheral nerve sheath tumour
Cell of origin	Myelinating Schwann cells (neural crest)	Non-myelinating Schwann cells	Cellular components of peripheral nerves, sacral or lumbar plexus
Setting	Paravertebral; symptoms of nerve compression	NF-1 (autosomal dominant) or sporadic; male; third–fifth decades	<10% of RP sarcomas; about half with predisposition (mostly NF-1) , arising from neurofibroma; sporadic cases linked to prior radiation ; in NF-1 often found in adolescence
Imaging, pathology	Homogeneous, circumscribed when small; cystic change and calcification with growth; encapsulated; S100 positive	Well circumscribed, homogeneous on CT	Radiologically indistinguishable ; fleshy, haemorrhage or necrosis; herringbone spindle cells; adjacent benign neurofibroma confounds small samples; S100 in about half, H3K27me3 loss
Presentation	Slow growing	—	Nerve irritation, pain, paraesthesia, paralysis
Treatment	Complete local resection usually curative	Surgical — malignant transformation up to 20% (NF-1, radiation exposure)	Biopsy is the gold standard for diagnosis; surgery for all

- **MPNST management:** preoperative chemotherapy and radiotherapy do not eradicate disease but **increase complete resection rates**. Infiltration of adjacent structures is uncommon, but **extension along nerve roots** occurs.
- Prognosis tracks **negative margins**; the parent nerve must be resected, so **neurological loss is inevitable**. **5-year survival about 70% with negative margins vs about 20% with positive margins**.
- Unresectable: **anthracycline-based chemotherapy and palliative radiotherapy**.

Concept — Neurogenic tumours are mostly benign and cured by excision; the exceptions — neuroblastic tumours in adults and MPNST — declare themselves by **systemic spread or nerve-root extension**, and histology alone rarely predicts which paraganglioma will metastasise.

HAEMATOLOGICAL CONDITIONS/LYMPHOMAS OF THE RETROPERITONEUM

- A significant share of RP masses. **Lymphoma is commonest**; extramedullary myeloma or plasmacytoma and other lymphoproliferative diseases also occur.

Lymphomas

- **Almost a third of primary RP tumours**. Hodgkin and non-Hodgkin types, with different prognoses.
- Circumscribed, **confluent, commonly homogeneous** masses along the RP and pelvic nodal chains.
- Key issue: **they resemble other RP tumours** → **FNA and/or core biopsy**; they **rarely, if ever, need surgery** and are treated with **chemotherapy**.

Plasmacytoma

- One spectrum: **focal = plasmacytoma; diffuse = multiple myeloma**. Myeloma is the commonest primary bone malignancy in adults; **extramedullary disease in <10% of new diagnoses** — abdomen/pelvis, skin and soft tissue, paraspinal regions; the RP is a rare site.
- **Serology:** raised immunoglobulin G (IgG); **serum electrophoresis "M spike"** — myeloma protein impairing immunity, coagulation and renal function.

- Diagnosis: **percutaneous FNA or core biopsy**; when suspected, **bone marrow aspirate and skeletal survey**. **Whole-body CT has replaced plain radiographs** for bone disease; **PET/CT assesses response**.
- Histology: monoclonal polygonal plasma cells, homogeneous amphophilic cytoplasm, eccentric nuclei; **CD138 and Bcl-2 positive**.
- Isolated RP disease: complete responses reported to **wide excision, radiotherapy and chemotherapy**.

Concept — Haematological RP masses are the group in which the **operation is the biopsy**: tissue diagnosis diverts the patient from resection to systemic therapy.

CYSTIC MASSES

- Purely cystic RP masses are rare: cystic lymphangioma, cystic teratoma, cystadenoma, cystadenocarcinoma, cystic mesothelioma, cystic change in solid tumours.
- Treatment differs greatly, so **diagnosis rests on presentation, imaging and serology** — because of general **caution against percutaneous biopsy or aspiration** (rupture risk, sampling limitations).

	Cystic lymphangioma	Mucinous cystadenoma	Cystic mesothelioma
Origin	Acquired loss or congenital abnormal communication of lymphatics with upstream channels	Unclear; possibly invagination of peritoneal mesothelium → mucinous metaplasia → occasional malignant transformation	Serous lining (pleura, pericardium, peritoneum); rare in RP; no association with asbestos
Who	Children mostly, also adults; male	Female	Female ; vague abdominal pain
Imaging	Thin-walled, multiseptated, crossing the midline into several compartments	Unilocular , homogeneous	Multilocular thin-walled cysts, like lymphangioma
Pathology	Cystic (commonest), cavernous, capillary; single endothelial layer; clear or milky lymph; CD31, CD34, D2-40	Single layer of columnar epithelium, basal nuclei, pale cytoplasm; aspiration cytology unhelpful	Calretinin, pan-keratin
Treatment	Complete excision ; aspiration and marsupialisation abandoned — high recurrence	Complete excision for diagnosis and treatment	Wide resection — local recurrence

Cystic Changes of Solid Retroperitoneal Tumours

- From serous or mucinous secretion, or **degeneration after chemotherapy or radiotherapy**. History is paramount: prior cancer treatment, constitutional symptoms, family history.
- Prior GCT → **cystic teratoma**, especially a **growing mass after chemotherapy or late relapse**.
- **Paraganglioma** can show central low attenuation (haemorrhage and liquefaction); similar change in **leiomyosarcoma, schwannoma and GIST**.

Retroperitoneal Cancer of Unknown Primary

- Up to **5% of cancers worldwide**; median age about **60**; slight male predominance. Evaluation rests on **radiological distribution and IHC** of biopsy tissue.
- Histology across all sites: **adenocarcinoma 50%, undifferentiated carcinoma 30%, squamous 15%**, remainder undifferentiated neoplasms. In the RP, **poorly differentiated carcinomas** predominate, often encasing the great vessels.
- Molecular: aberrations of **chromosomes 1 and 12**; Bcl-2 and p53 overexpression in up to half; increased c-myc, ras and c-erbB-2.

- **Young man → test for isochromosome 12p**, which identifies an RP primary GCT; markers: **AFP/β-hCG in young men, prostate-specific antigen (PSA) in older men.**
- Primary identified in **fewer than 20%**. Whole-body imaging for staging and primary search: CT the mainstay, PET-CT increasingly used.

Presentation	Initial therapy
Central/midline RP, especially i(12p) positive	Platinum-based combination chemotherapy ; response in up to half, complete radiological response in 15–25%
Poorly differentiated neuroendocrine carcinoma	Platinum-based
Single, small, localised deposit	Surgical extirpation and/or focused radiotherapy

- Future: gene-expression profiling and whole-genome analysis to assign tissue of origin and individualise therapy.

KEY POINTS

- Many histological entities in the RP present as an infiltrative or expanding mass.
- A multidisciplinary approach — radiological staging, pretreatment biopsy when indicated, serological and IHC characterisation, and multimodal therapy (surgery, systemic therapy, radiotherapy) — optimises outcomes.

Concept — *In the RP, histology decides the first treatment more than anatomy does: chemotherapy-first entities (GCT, lymphoma, Ewing sarcoma, midline cancer of unknown primary) must be excluded before a sarcoma resection is planned, and cystic lesions are characterised without a needle.*

SURGICAL MANAGEMENT OF THE RETROPERITONEAL TUMOUR

Preoperative Considerations

- **No bowel preparation** or dietary modification.
- **GCT: serum tumour markers within 7–10 days** of surgery.
- Extra **blood products** for complex resections.
- **Sperm banking** for men wanting future paternity when RP masses lie in the path of the **postganglionic sympathetic fibres**.
- **Bleomycin**: a medical oncology partner assesses toxicity, limits dose and obtains **pulmonary function tests** to reduce postoperative **acute respiratory distress syndrome**; the anaesthetist must know of prior bleomycin — **low fraction of inspired oxygen (FiO₂) and conservative fluid resuscitation**.
- **Imaging**: CT abdomen and pelvis reviewed at consultation and again immediately before surgery; **current chest CT** if prior pulmonary masses, planned thoracic resection, or radiological/serological progression. **Imaging within 6 weeks of surgery**. Careful review usually prevents unplanned intraoperative consultations.
- **Total IVC thrombosis** identified preoperatively simplifies the operation — the IVC is resected. **Incomplete occlusion** requiring resection may need reconstruction with **cadaveric allograft or ringed polytetrafluoroethylene (PTFE) graft**, provided there is no bowel contamination.

Surgical Technique

- **Orogastric tube** suffices; **nasogastric tube** only for duodenal resection/repair or high-volume masses needing complete mesenteric mobilisation with bowel on the chest.

- Supine → **ventral midline incision** → inspect abdominal viscera → **falciform ligament ligated and divided** (prevents hepatic retraction injury) → self-retaining retractor.

Exposure of the Retroperitoneum

Mass	Exposure
Smaller paracaval and interaortocaval	Open root of mesentery from inferior tip of caecum to medial aspect of inferior mesenteric vein (IMV)
Large right, interaortocaval and/or paracaval	Continue around inferior caecum to right white line of Toldt, up to the foramen of Winslow → bowel on chest
Larger left and para-aortic	IMV ligated and divided
Small-volume left	Through the left white line of Toldt

- Plane between mesentery and RP fat developed **along the anterior surface of the gonadal vein** → duodenum dissected off IVC and left renal vein → **superior mesenteric artery (SMA) identified by palpation** → retractor blades placed **either side of the SMA**.

APPROACH TO THE RETROPERITONEUM

Split-and-Roll Technique

- The IVC and aorta are central, so every approach is organised around exposing them. **Large lymphatics over the left renal vein are ligated and divided.**

ORDER OF THE SPLIT WHEN BOTH GREAT VESSELS ARE SPLIT

Aorta first (authors' preference)	IVC first
Avoids injury to precaval right accessory lower-pole renal arteries	Right postganglionic sympathetic fibres are identified and traced to the superior hypogastric plexus , protecting them during the aortic split

- **Aortic split:** start at **12 o'clock just below the left renal vein** → continue caudally → identify the IMA prospectively: **preserve it in a right-sided procedure; doubly ligate and divide it in a bilateral dissection** to expose the left para-aortic region → if nerve-sparing, **stop at the IMA** and identify postganglionic fibres before going caudally.

Left Para-aortic Region

- Approach laterally (left white line of Toldt) or medially (mesenteric root) by disease distribution. Sequence: left gonadal vein doubly ligated where it crosses the ureter → **ureter swept laterally behind a retractor** → split continued at 12 o'clock on aorta and left common iliac artery to the ureter → tissue rolled laterally off aorta and iliac → **three left lumbar arteries** (renal hilum to bifurcation) doubly ligated and divided → mass rolled inferiorly off the left renal vein; left gonadal and lumbar veins ligated at their entry into it → packet freed from lower pole of kidney and ureter → caudal packet rolled superiorly off body wall, **preserving left genitofemoral nerve and sympathetic trunk**, controlling lumbar veins and arterial stumps → rolled up to the crus; **retrocrural lymphatics ligated** → **release ureteric retractor** to prevent prolonged ischaemia.

Interaortocaval Region

- **Right nerve-sparing** → **IVC split first**; otherwise control the medial side of the aorta first. Sequence: IVC split from renal hilum to the crossing of the right common iliac artery, then inferolaterally to the right ureter → right gonadal vein ligated at the IVC → tissue rolled medially off IVC, **nerves seen running obliquely along the lateral edge of the packet** → **lumbar veins** (renal hilum to common iliac veins) ligated — **number and position unpredictable, unlike the arteries** → tissue rolled laterally off IVC →

right gonadal vein ligated where it crosses the ureter; ureter retracted → tissue rolled medially off aorta; **medial three lumbar arteries** divided → packet harvested off the anterior spinous ligament, **preserving the right sympathetic trunk** at its right lateral border and controlling lumbar stumps → superior packet rolled inferiorly off renal vessels to the crus → **retrocrural lymphatics ligated** (avoiding the renal artery) to prevent lymph leak and **chylous ascites**.

Right Paracaval Region

- **Smallest of the three packets** — right kidney and ureter lie close to the IVC. Rolled laterally and superiorly off the right common iliac artery to the ureteric crossing → superiorly off psoas fascia, **preserving right sympathetic trunk and genitofemoral nerve** → towards right renal hilum and crus; the packet often **tapers and passes under the IVC** before the hilum.

Concept — *Split-and-roll converts a mass into **vessel-defined packets**: split each great vessel at 12 o'clock, roll tissue off its sides, and control the edges — lumbar vessels, sympathetic fibres and retrocrural lymphatics — where the complications arise.*

SURGICAL MODIFICATIONS

Template Considerations for Management of Metastatic Germ Cell Tumours

Primary Retroperitoneal Lymph Node Dissection

- In the 1970s and 1980s three developments reshaped RP management of testicular GCT: **curative cisplatin-based chemotherapy**; recognition of **distinct lymphatic spread for right- versus left-sided tumours**; and **nerve-sparing techniques** preserving the postganglionic fibres for seminal emission and antegrade ejaculation.

LANDMARK TEMPLATE STUDIES

Study	Design	Finding and contribution
Ray et al., 1974 (Memorial Sloan Kettering Cancer Center (MSKCC))	283 RPLNDs, 1944–1971; mainly infrahilar	Evolved from full bilateral to " modified bilateral " as primary landing zones became apparent; like unilateral templates but tissue below the IMA routinely resected . First side-specific templates
Donohue et al., 1982 (Indiana)	Mapping in 104 pN+ patients; full bilateral including suprahilar in all	Left → left para-aortic; right → interaortocaval and precaval . Contralateral and suprahilar spread rare, rising with tumour bulk ; interiliac rare. Validated modified bilateral templates for low-stage disease
Weissbach and Boedefeld, 1987	Multi-institutional, 214 non-bulky pathological stage (PS) II	Reduced left template: para-aortic plus upper pre-aortic; frozen section of primary landing zone — if positive, full bilateral infrahilar RPLND
Eggerer et al., 2007 (MSKCC)	500 primary RPLNDs for clinical stage (CS) I/IIA; five modified templates applied to PS II packets	3–23% of pN+ patients had disease outside a modified unilateral template , more with right-sided tumours → recommended full bilateral infrahilar nerve-sparing RPLND for CS I/IIA

- **Omitting contralateral and interiliac dissection preserves antegrade ejaculation** in most; **omitting suprahilar dissection reduces chylous ascites, renovascular injury and pancreatic complications**.
- **Consensus:** suprahilar/retrocrural and interiliac resection can be omitted from the standard template. **Controversy:** whether the contralateral RP must be resected.
- No study has compared modified unilateral with full bilateral templates. **Cancer-specific survival (CSS) and OS approach 100%** in all series, so wider templates cannot improve them; the real question is

whether a full bilateral infrahilar dissection prevents **RP relapses** after a properly performed unilateral one. Series from centres using each approach show **similar outcomes**.

TABLE 140.2 SELECTED PRIMARY RPLND SERIES (REBUILT)

Centre and approach	pN+ rate	Relapse, pN0	Relapse, pN+ after RPLND alone	CSS
Indiana (Donohue 1993, Hermans 2000, Douglass 2020) — predominantly modified unilateral in the contemporary series	About a quarter to under half	About 8–12%	About 19–34%	99–100%
MSKCC (Stephenson 2005) — bilateral infrahilar	About 30%	About 7%	About 34%	99.7%
Nicolai 2004	20%	Not reported	Not reported	98.8%

Follow-up about 4–7 years across series.

- **What the table is for:** showing that template choice does not move survival. **Pattern:** CSS is 99–100% whatever the template; relapse rates differ mainly with adjuvant policy.
- **Trap:** MSKCC's higher "cure by surgery alone" reflects **routine adjuvant chemotherapy for pN2**; Indiana randomised node-positive patients to observation or adjuvant chemotherapy (1993), later observed pN1 and most pN2, reserving chemotherapy for relapse and pN3 (2000). The contemporary Indiana series, almost entirely modified unilateral, achieved near-100% CSS and OS with only a small minority receiving adjuvant chemotherapy.
- The appropriate template remains controversial; the Ray, Donohue, Weissbach and Eggener templates all give excellent survival, and which best balances oncological control against morbidity is **unresolved**.

Concept — With survival at a ceiling, the template debate is about **morbidity versus in-field relapse**; lymphatic spread is side-specific and predictable, and bulk is what breaks the rule.

Postchemotherapy Retroperitoneal Lymph Node Dissection

- After cisplatin chemotherapy, most teratoma and viable tumour lie in the primary landing zones, but **bulky disease crosses over** and histology cannot be confirmed intraoperatively → postchemotherapy (PC)-RPLND "as complete as possible".
- **Standard PC-RPLND: resection of all macroscopic disease plus full bilateral infrahilar dissection** — excellent local control but significant morbidity, including **anejaculation** when nerves cannot be spared.
- With bilateral dissection, disease lies **outside a unilateral template in 18–32%**, but outside both the template and the macroscopic mass in only **2–18.6%** — variation reflects selection and template.

PRIMARY VERSUS POSTCHEMOTHERAPY RPLND

	Primary RPLND	PC-RPLND
Standard template	Modified unilateral or full bilateral infrahilar (debated)	All macroscopic disease + full bilateral infrahilar
Unilateral template acceptable when	Low-stage disease in primary landing zone	Only with strict selection criteria (below)

Criteria for modified unilateral PC-RPLND:

1. **Well-defined lesion ≤5 cm confined to the primary landing zone** on imaging before and after chemotherapy.
2. **Normal postchemotherapy serum tumour markers.**

3. IGCCCG good or intermediate risk.

- With these criteria: **in-field RP recurrence 0–1%, antegrade ejaculation 85–99%, CSS 98–100%**. No prospective comparison exists; bilateral remains the standard of care, and unilateral templates require **strict adherence** to the criteria.

Nerve-Sparing Retroperitoneal Lymph Node Dissection

- **Four postganglionic efferent sympathetic fibres (L1–L4)** mediate antegrade ejaculation; anatomy varies widely. **L2 and L3 are usually fused**. L2–L4 run anteriorly along the aorta and common iliac vessels; **L1 runs a shallower, more caudal, oblique course**, leaving the sympathetic trunk near the ipsilateral renal hilum.
- **Left fibres:** found along the lateral border of the aorta and left common iliac artery and onto their anterior surface **immediately caudal to the IMA** → Kittner sponge sweeps fat to reveal shiny, off-white oblique fibres joining the contralateral fibres in the **superior hypogastric plexus** → vessel loops for gentle traction → traced to origin on the trunk. Alternatively, find the left trunk distal to the IMA and trace cranially.
- **Right fibres:** seen as precaval and interaortocaval tissue is **rolled medially off the IVC**, running obliquely anteriorly and inferiorly to the superior hypogastric plexus → cleared, looped and traced to the right trunk.
- Then dissect the packets around the fibres, **passing the specimen through the web of fibres** as it is released from the body wall. Fibres leave the trunk **close to lumbar vessels** → at risk when controlling lumbar bleeding.
- After bilateral nerve-sparing, or unilateral nerve-sparing with contralateral complete resection, a **temporary anejaculatory period** may occur — probably **neuropraxia**.

AUXILIARY PROCEDURES

- Relevant to **advanced, bulky RP disease**; rarely needed in low-volume disease. **Nephrectomy is commonest, then vascular resection or reconstruction**; the likelihood of adjacent-organ resection rises with RP tumour volume.

Nephrectomy

- **Commonest auxiliary procedure at PC-RPLND:** 5–31%; in primary RP sarcoma, **up to 55%**.
- **Risk factors:** high-risk settings — **salvage, desperation, late-relapse or reoperative RPLND**; **RP mass size** (mass >10 cm carries far higher odds than <2 cm); **left primary with left para-aortic mass**.
- Renal function declines after nephrectomy, but **without need for renal replacement therapy and without compromising later chemotherapy**; progression to chronic kidney disease depends on **age and preoperative glomerular filtration rate**.

Vascular Reconstruction

AORTIC VERSUS CAVAL RESECTION

	Aortic replacement	Vena caval resection
Frequency	About 1% of PC-RPLND	5–10% of GCT surgery
Indication	Tumour fixed to aorta ; most had salvage chemotherapy and/or raised markers	Tumour clearance, caval scar occlusion, caval tumour thrombus (commonest) ; most specimens contain viable cancer or teratoma
Reconstruction	Always — aortic tube graft, or aortobi-iliac graft if iliacs involved	Not routinely required

	Aortic replacement	Vena caval resection
Outcome	Worthwhile: a meaningful minority disease-free at medium-term follow-up, given chemoresistant bulky disease	Most have no or mild long-term disability ; higher risk of chylous ascites and perioperative complications

Aortic Replacement

- **Anticipate preoperatively** — alert vascular surgery, counsel the patient. Sequence: IVC dissected off mass and aorta by split-and-roll with lumbar vein division → left ureter freed → left renal hilum freed if not encroached → vascular team secures **adequate aorta cranial and caudal** to the tumour → **intravenous heparin before cross-clamping** → aorta cross-clamped and resected **en bloc** with the mass, lumbar arteries divided → graft sewn by standard vascular principles.

Vena Caval Resection

- Usually bulky disease. **Lower-limb oedema and imaging suggesting IVC compression/occlusion → cavography, US or MRI** to assess flow and guide intraoperative decisions.
- Why caval ligation is tolerated in testis cancer: **slow tumour growth and desmoplastic reaction to chemotherapy occlude the IVC gradually**, allowing **venous collaterals** — morbidity is lower than after acute occlusion, especially when complete occlusion with collaterals exists preoperatively.
- **Sarcoma**: caval resection ± reconstruction for negative margins; grafts (Dacron, PTFE, homograft) **thrombose in 10–40%**; anticoagulation varies — consider **at least 3 months**.

Tumour Thrombectomy

- Intraluminal thrombi are infrequent in GCT and sarcoma: **vena cava commonest**, then renal vein, aorta least.
- Caval thrombi → cavectomy, partial cavectomy or thrombectomy; **renal vein thrombi → nephrectomy** in nearly all.
- Histology: bland thrombus, necrosis, teratoma, residual cancer — **nearly half contain teratoma or viable cancer**.

Concept — *The chronicity of the obstruction decides reconstruction: a **gradually occluded IVC** has **collaterals and can be ligated**, whereas the aorta carries obligatory flow and must always be replaced.*

COMPLICATIONS

Ejaculatory Function

- **Modified unilateral template without nerve-sparing**: early series 75–87% antegrade ejaculation; contemporary **97%**; **with ipsilateral nerve-sparing 99%** — gains from avoiding contralateral fibres **caudal to the IMA**.
- **Nerve-sparing RPLND**: antegrade ejaculation **90–100%**.
- Temporary anejaculation after bilateral templates reflects **neuropraxia**; it was not seen when unilateral templates with ipsilateral nerve-sparing were used.

Vascular

- Major vascular resection modestly increases vascular morbidity: after caval reconstruction, **perioperative thrombosis in about one in five** (deep vein thrombosis and pulmonary embolism).
- **Venous thromboembolism after RPLND is low** (young, healthy patients): **pulmonary embolism <1% after primary RPLND, 0.1–3.1% after PC-RPLND**; deep vein thrombosis 0–1% (under-reported, often asymptomatic).

Lymphatic

	Lymphocele	Chylous ascites
Definition	Lymph collection in the resection bed	Chylomicron-containing lymph in the peritoneal cavity
Incidence	Subclinical common; symptomatic 0–1.7%	Primary RPLND 0.2–2.1%; PC-RPLND 2–7%
Presentation	Ureteric compression, visceral displacement (if large), infection	Abdominal fullness, anorexia, nausea, vomiting, pain, dyspnoea; fluid wave (distinguishes from ileus); weight gain
Diagnosis	CT: thin-walled cyst in resection bed; air or rim enhancement → infection	Milky fluid; alkaline, Sudan black positive, triglycerides above serum — usually unnecessary; clinical and gross inspection suffice
Prevention	Ligate large-calibre lymphatics	Ligate retrocrural lymphatics; avoid needless suprahilar dissection
Treatment	Percutaneous drainage ; antibiotics only if infected; if infected, leave an indwelling drain rather than aspirate	Graduated (below)

Chylous Ascites — Risk Factors

- **Suprahilar resection** — disruption of the **cisterna chyli (L1–L2, medial to the posterior aorta, retrocrural)** and its tributaries.
- **IVC resection** — raised infracaval venous pressure → capillary leak and third-spacing of lymph.
- **More preoperative chemotherapy cycles, greater blood loss, longer operation.**
- **Trap:** left suprahilar dissection can injure the **pancreatic tail**, also causing large-volume ascites → check drained fluid for **amylase and lipase**.

Chylous Ascites — Graduated Management

- **Paracentesis + low-fat/medium-chain triglyceride (MCT) diet ± intramuscular octreotide** → reaccumulation: **indwelling drain** → if dietary measures already in place: **nil by mouth + total parenteral nutrition (TPN)** → persistent drainage **>100 mL/24 h** (exceedingly rare): continued conservative management, **peritoneovenous (LeVeen) shunt**, or **surgical ligation of the leak** — the last two are **last resorts** (shunt occlusion or malfunction needing revision, sepsis, fat embolism) → after resolution, continue **low-fat MCT diet for 1–3 months**.
- Octreotide is unstudied in urology but reduces chyle leak after hepaticopancreaticobiliary surgery.

Neurological

- After PC-RPLND, **paraplegia is not seen**; peripheral nerve injuries arise from **positioning and retractor placement (femoral neuropraxia)** — careful positioning prevents them.
- **Paraplegia** occurs after resection of **bulky combined mediastinal and RP disease**; neurological risk rises with the extent of para-aortic resection.
- **Primary RP sarcoma:** about **5% need major lumbar nerve resection**, usually because the sarcoma arises from the nerve root. Commonest: **femoral and obturator**; sensory: genitofemoral, ilioinguinal. **Femoral nerve loss** → **knee extension weakness**, sometimes needing a **hinged knee brace**.

Postoperative Care

- **Drains not routine**; a Jackson-Pratt drain for large-volume RP, retrocrural or duodenal resections, removed once on a **regular diet with serous drainage <100 mL/24 h**.
- Without bowel repair: ice chips evening of surgery → **day 1** unlimited clear fluids, chair and ambulation → **day 2** regular diet, off intravenous analgesia → discharge **days 2–5**; longer after larger resections.

KEY POINTS

- Surgery for an RP mass requires consideration of all risks and obstacles and a planned approach to exposure and mobilisation.
- Advance preparation and coordination of consultation for possible auxiliary procedures are imperative.
- The surgeon should know, and counsel patients on, the short- and long-term complications of RP surgery.

Concept — Most complications of RP surgery are **functions of extent**: *suprahilar and caval work drive chylous ascites, bilateral and caudal-to-IMA dissection drives anejaculation, and para-aortic and nerve-root resection drive neurological loss.*

DISEASE-SPECIFIC NEOADJUVANT AND ADJUVANT THERAPIES

Radiation

- Use of radiotherapy with RP sarcoma resection has grown, aiming to **reduce local recurrence**, extrapolated from extremity sarcoma. Timing, dose and field remain debated; long-term toxicity and recurrence patterns need study.

EVIDENCE FOR PERIOPERATIVE RADIOTHERAPY IN RP SARCOMA

Study	Comparison	Verdict
Early randomised trial	Intraoperative + postoperative external beam vs postoperative external beam alone	Fewer locoregional recurrences with intraoperative arm, similar OS ; toxicity heavy in both — enteritis more with external beam alone, peripheral neuropathy more with intraoperative ; adjunctive chemotherapy (cyclophosphamide, doxorubicin, methotrexate) added nothing
Phase I/II	Low-dose brachytherapy + preoperative radiotherapy	Closed for severe toxicity
EORTC-62092 STRASS (phase III)	Preoperative radiotherapy + surgery vs surgery alone ; 266 patients	Abdominal recurrence-free survival no different (hazard ratio (HR) 1.01) ; more complications with radiotherapy → not standard of care
National Cancer Database	Preoperative or postoperative vs none (propensity-matched)	Survival advantage with any radiotherapy
Surveillance, Epidemiology, and End Results (SEER)	Adjuvant vs none	Conflicting : one analysis favoured adjuvant radiotherapy; a larger one found no difference except in MFH
Intraoperative radiotherapy (IORT) series	IORT, cumulative dose >60 Gy	No effect on 5-year local recurrence

- Retrospective studies cannot assess local control, toxicity, dose, timing or selection bias.
- **Late toxicity at 50–60 Gy** is mainly **bowel**, mostly with **adjuvant** rather than preoperative radiotherapy.
- **Bottom line**: radiotherapy improves local control at the cost of toxicity (greatest postoperatively) with marginal OS gain; if used, **preoperative** is most promising, but STRASS tempers enthusiasm. Intraoperative electron beam and brachytherapy remain **investigational**; a phase I/II trial is testing neoadjuvant radiotherapy then surgery with IORT in high-risk RP sarcoma.

Chemotherapy and Systemic Therapies

- Outside platinum chemotherapy for metastatic GCT, data for systemic therapy in **localised RP sarcoma are limited**; the commonest RP sarcomas — **liposarcoma and leiomyosarcoma** — are largely **chemoresistant**. Some histologies are treated primarily by chemotherapy (lymphoma), so **accurate characterisation is pivotal**.
- In non-metastatic disease the question is whether neoadjuvant therapy reduces distant spread, improves resectability and reduces positive margins.
- **Responsive histologies: myxoid liposarcoma, synovial sarcoma, Ewing sarcoma**. Standard neoadjuvant chemotherapy is recommended **only for Ewing sarcoma**.
- **Resectable tumour**: neoadjuvant benefit uncertain. **Resectability in doubt, or regional/distant spread: consider multimodal therapy with induction chemotherapy**.
- **Minimally responsive tumours** (WD liposarcoma, leiomyosarcoma): resect local, regional and/or metastatic disease when feasible in selected cases; otherwise **clinical trials**.
- Early successes: **MDM2 and CDK4 inhibitors** in liposarcoma; **tyrosine kinase inhibitors** in refractory soft-tissue sarcoma; immune-modulating therapy.
- **STRASS II** (ongoing): neoadjuvant doxorubicin + ifosfamide (leiomyosarcoma: **doxorubicin + dacarbazine**) plus surgery vs surgery alone.

TABLE 140.3 CHEMOTHERAPY TRIALS IN SARCOMAS (REBUILT BY HISTOLOGY)

Histology	Setting	Regimen	Result
RP sarcoma, all types	Neoadjuvant (minority) vs none — large database and multicentre series	Not reported	No OS improvement ; one database showed worse survival with chemotherapy; no benefit of preoperative chemotherapy or radiotherapy
Soft-tissue sarcoma, metastatic	Database, RP a minority	Not reported	Survival longer with chemotherapy overall, but by histology benefit confined to UPS
Liposarcoma, metastatic	Retrospective series	Doxorubicin/ ifosfamide; anthracycline-based	Response: WD none, DD modest, myxoid best ; mostly stable disease, complete response rare; WD vs DD response not clearly different in one series
Myxoid liposarcoma	Mostly neoadjuvant	Doxorubicin/ ifosfamide	Partial responses in over a third, remainder stable; high R0 rate, all R0 cases also had preoperative radiotherapy
Synovial sarcoma	Neoadjuvant and metastatic	—	Better PFS than other histologies
MPNST	Neoadjuvant, single RP patient	Ifosfamide, epirubicin	Response and R0 in that patient; data minimal

- **What the table is for**: showing that "sarcoma" does not predict chemosensitivity — histology does. **Pattern**: pooled RP sarcoma series show no benefit because the pool is dominated by chemoresistant liposarcoma and leiomyosarcoma.
- **Trap**: the negative pooled result does not apply to **myxoid liposarcoma, synovial sarcoma or Ewing sarcoma**.

KEY POINTS

- Adjuvant and neoadjuvant therapies will keep expanding within multimodal care; testicular cancer is the exemplar.

- In sarcoma, histological heterogeneity and differing responses to radiotherapy and systemic therapy prevent sweeping recommendations on timing and sequencing.
- A multidisciplinary team exploring all options gives the best chance of success.

Concept — Testicular GCT is the model of **protocolised multimodal therapy**; RP sarcoma is its opposite — adjuncts are histology-specific, largely unproven, and surgery with negative margins remains the backbone.

CONCLUSIONS

- The RP mass needs careful initial characterisation because some entities follow **divergent, non-surgical pathways**.
- **Metastatic GCT**: standardised pathways; **refer to experienced centres**. **RP sarcoma**: likewise, given rarity, variable behaviour and multidisciplinary needs; the role of neoadjuvant or adjuvant chemotherapy or radiotherapy is limited by data.
- As most RP lesions ultimately need surgery, familiarity with their clinical, radiological and histological features and perioperative needs is essential.